

- 1) Explain the relationship between risk and return.

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- 2) What are diversifiable and non-diversifiable risks?

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18.4 VALUATION OF PORTFOLIO AND EXPECTED RETURNS FROM A PORTFOLIO

As we have discussed earlier, a portfolio is a collection of financial assets like stocks, bonds, mutual funds, exchange-traded funds (ETFs), real estate, commodities, and other investment vehicles owned by an individual, institution, or entity. The purpose of building a portfolio is to diversify risk and potentially maximize returns by spreading investments across different asset classes, industries, and geographic regions. The expected return and valuation of the Portfolio act as guiding principles and significantly influence investment choices and the assessment of performance.

18.4.1 Valuation of Portfolio

Portfolio valuation entails evaluating the value of an entire investment portfolio at a point in time. This process includes assessing the worth of each asset within the portfolio, taking into account various factors like market prices, potential earnings, and intrinsic value. Precise portfolio valuation is essential for evaluating performance, making well-founded investment choices, and ensuring transparency with stakeholders.

Let a portfolio be a collection of n assets, each with share price P_i ($i=1,2,3,\dots,n$) Let there be N_i shares of each asset i .

Then, the total value of the portfolio will be:

$$V = N_1P_1 + N_2P_2 + N_3P_3 + \dots \dots N_nP_n = \sum_{i=1}^n N_iP_i$$

A typical portfolio has $V > 0$ with the weight of each investment as:

$$w_i = \frac{N_iP_i}{N_1P_1 + N_2P_2 + \dots \dots N_nP_n} = \frac{N_iP_i}{V}$$

Factors affecting portfolio valuation:

- 1) *Asset Evaluation*: Determining the value of individual assets within a portfolio involves the valuation of each asset class. Some of the evaluation methods are discounted cash flow analysis, price-to-earnings ratios, or comparative company analysis.
- 2) *Market Dynamics*: Fluctuations in market conditions and liquidity constraints can impact asset prices, thereby influencing portfolio valuation. Regular monitoring of market trends ensures that portfolio valuations accurately reflect prevailing market conditions.
- 3) *Impact of Diversification*: The benefits of diversification inherent in a portfolio can affect its valuation. Diversified portfolios may display reduced volatility and higher valuation metrics owing to the mitigation of risk.

18.4.2 Expected Return and Risk from a Portfolio

The portfolio return is a weighted average of the individual returns.

Consider a portfolio of n assets with weights: $\{w_1, w_2, w_3, w_4 \dots w_n\}$, $\sum_{i=1}^n w_i = 1$

Then, the return on the portfolio will be:

$$\tilde{r}_p = w_1\tilde{r}_1 + w_2\tilde{r}_2 + \dots \dots w_n\tilde{r}_n$$

Where w_i is the weight of each asset and $\tilde{r}_i$ is a return from each investment

The expected return on the portfolio is:

$$\bar{r}_p = E[r_p] = w_1\bar{r}_1 + w_2\bar{r}_2 + \dots w_n\bar{r}_n$$

Where w_i is the weight of each asset and $\bar{r}_i$ is the expected return from each investment

Risk

The variance and standard deviation are used as a measure of risk. According to modern portfolio theory, unlike the expected return, the variance of a portfolio is not a simple weighted average of the individual variances. Rather, it is affected by the correlation coefficient of securities also.

The general form of the relationship for an n security portfolio is:

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n x_i x_j \rho_{ij} \sigma_i \sigma_j$$

Where σ_p^2 = portfolio variance

x_i = proportion of total investment in asset i

x_j = proportion of total investment in asset j

ρ_{ij} = correlation in the returns of securities i and j

σ_i = security standard deviation

Let us understand it with the help of a two-security case:

$$\sigma_p^2 = x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2x_A x_B \rho_{AB} \sigma_A \sigma_B$$

All else being equal, risk is something people want to avoid. One measure of risk is portfolio variance. It represents total risk. Intuitively, most investors want portfolio variance to be as low as possible without having to give up any return. In the above equation, the total risk (σ_p^2) has three components: one from the variance of stock A, another from the variance of stock B, and a third from the relationship between returns of Stock A and Stock B i.e., interactive risk.

As the number of securities in the portfolio increases, the number of terms on the right-hand side of the equation increases as well. With four securities, there will be four variance components and eight interaction components from the covariance matrix.

The portfolio manager does not have any control over individual securities but has control over the selection of portfolios. They can try to choose securities which have the least correlation between them so that interaction terms can be reduced and total risk be minimised.

18.5 MARKOWITZ PORTFOLIO THEORY

Traditionally, portfolio theory focused on analyzing individual securities while considering the balance between risk and return. This theory operated on the assumption that securities should be selected based on minimizing risk, measured by their standard deviation from the mean of expected returns. According to this approach, there's a direct relationship between the variability of returns and risk: the higher the variability, the higher the risk, and vice versa. Therefore, investors typically opt for assets with the lowest variability of returns. To determine the return on an individual security, factors such as the company's dividend payments, price-earnings ratio, the duration of holding, and the market value of shares are considered.

Modern Portfolio Theory (MPT), a hypothesis put forth by Harry Markowitz in his paper "Portfolio Selection," (published in 1952 by the Journal of Finance) opened the door to modern portfolio theory. It is an investment theory based on the idea that risk-averse investors can construct portfolios to optimize or maximize expected return based on a given level of market risk, emphasizing that risk is an inherent part of higher reward. The modern portfolio theory emphasises the need for maximization of returns through a combination of securities whose total variability is lower. It is not necessary that the success could be achieved by trying to get all securities of minimum

risk. This was the first major publication indicating the importance of security return correlation in the construction of a stock portfolio. Markowitz showed that for a given level of expected return and a given security universe. (The entire collection of alternatives from which specific investment choices are made), finding the specific portfolio that dominates the others requires knowledge of the covariance or correlation matrix between all possible security combinations.

After the publication of this paper, numerous investment firms and portfolio managers began to program Markowitz algorithms, which prescribed portfolio proportions to minimize portfolio variants. Even today, the term Markowitz Diversification refers to portfolio construction accomplished using security covariances.

18.5.1 Assumptions of the Markowitz Model

- 1) Investors are rational.
- 2) The supply and demand equilibrium is instantly achieved.
- 3) There are no arbitrage opportunities.
- 4) Price moves are efficient
- 5) Access to information is available to all participants.
- 6) The market is liquid.
- 7) There is no transaction cost.
- 8) There are no taxes.
- 9) Everyone has the same opportunity of borrowing and lending.

The model is also known as the mean-variance model because it is based on the expected returns and the standard deviation of different portfolios. It helps to make the most efficient selection by analysing various portfolios of the given assets. It shows investors how to reduce the risk in case they have chosen assets, not moving together.

Under the assumptions, a single asset or portfolio of assets is considered to be efficient. If no other asset or portfolio of assets offers higher expected returns with the same or lower risk or lower risk with the same or higher expected Return.

18.5.2 Opportunities set/ Feasible set of Portfolios

With a limited number of securities, an investor can create numerous portfolios by combining these securities in various proportions. These combinations form what's called the portfolio opportunities set, representing the feasible options for investment. Each portfolio in this set is defined by its

expected return and a measure of risk, typically variance or standard deviation of returns.

Not every portfolio in the opportunities set is appealing to an investor. Some portfolios are superior to others. A portfolio dominates another if it either offers a lower standard deviation with the same expected return, or a higher expected return with the same standard deviation. Inefficient portfolios, which are dominated by others, are not of interest to investors. They focus instead on efficient portfolios, which offer the best risk-return trade-off.

18.5.3 Efficient Portfolio Frontier

Harry Markowitz initially conceptualized portfolio theory by defining the objective as identifying a collection of efficient portfolios. An efficient portfolio is one where no higher expected return can be attained without accepting a higher level of risk. Conversely, it's a portfolio where no greater certainty of returns can be obtained without sacrificing expected returns. Once this efficient set of portfolios is determined at the asset class level, the investment manager selects an allocation along that frontier that aligns most closely with the investor's desired risk-return profile for the fund. This collection of efficient portfolios is commonly referred to as the efficient frontier. Figure 18.1 illustrates a typical representation of such efficient portfolios.

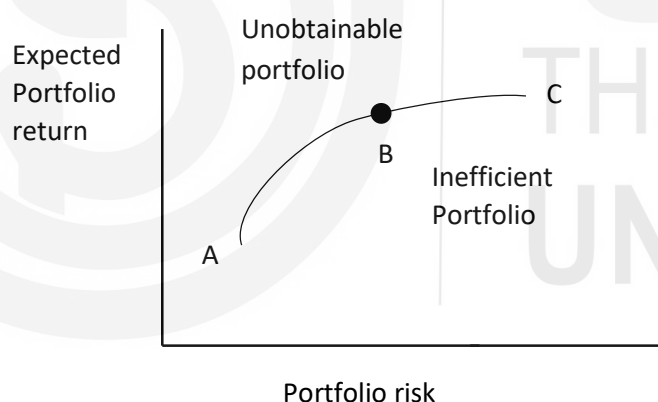


Fig. 18.1: Efficient Portfolio Frontier

This chart depicts the relationship between the expected return and the level of risk associated with a portfolio, or conversely, the risk of a portfolio about its expected return. The curved line shown in Figure 18.1 represents the efficient frontier, outlining the boundary of all possible portfolios formed by various combinations of selected asset classes. Portfolios lying inside this frontier are deemed inefficient because they offer less return for a given level of risk. Conversely, portfolios lying outside the line are more desirable, but they are not feasible to achieve.

Portfolio A represents the portfolio with the minimum risk, providing the least amount of risk among all possible combinations of asset classes. On the other hand, Portfolio C represents the portfolio with the maximum return and

the maximum risk. Although Portfolio C entails a 100% investment in the asset class offering the highest return, Portfolio A typically won't consist solely of the least risky asset class. This is because diversification across asset classes with varying risks helps mitigate overall risk. For portfolios seeking a balance between return and risk, options such as Portfolio B, positioned away from the endpoints, are usually preferred.

18.5.4 Criticisms of Markowitz's Portfolio Theory

While Markowitz's portfolio theory provides a valuable framework for portfolio construction, it is not without its limitations and criticisms. Some of the criticisms are:

- 1) **Unrealistic assumptions:** Markowitz's theory relies on several assumptions that might not hold in real-world scenarios. For instance, it assumes that investors are rational, risk-averse, and have access to unlimited borrowing and lending at a risk-free rate. However, in reality, investors may not always behave rationally, and their risk preferences can vary.
- 2) **Complexity and Practicality:** Implementing Markowitz's theory can be challenging due to its complexity. It involves intricate calculations and assumptions about correlations and expected returns, which may not always be accurate or practical to estimate.
- 3) **Sensitivity to Input Data:** The efficiency of portfolios generated by Markowitz's theory is highly sensitive to input data such as expected returns and covariance matrices. Small changes in these inputs can lead to significantly different portfolio allocations, making the results unstable and difficult to rely on.
- 4) **Neglect of Transaction Costs and Taxes:** Markowitz's theory assumes there are no transaction costs and taxes associated with portfolio rebalancing. In real-world investing, these costs can significantly impact returns and portfolio performance.
- 5) **Single-Period Analysis:** The theory primarily focuses on single-period analysis, neglecting the dynamic nature of markets and investors' changing risk preferences over time. This limitation can lead to suboptimal portfolio decisions in dynamic market conditions.
- 6) **Ignorance of Behavioural Finance Insights:** Markowitz's theory largely ignores insights from behavioural finance, which studies how psychological biases influence investor behaviour. Behavioural factors such as herd mentality, overconfidence, and loss aversion can significantly affect portfolio decisions but are not adequately accounted for in traditional portfolio optimization frameworks.
- 7) **Inability to Capture Tail Risks:** Traditional portfolio optimization techniques, including Markowitz's theory, often fail to adequately

capture extreme events or tail risks, leading to underestimation of potential portfolio losses during market downturns.

Check Your Progress 3

- 1) What are the key points of Markowitz's portfolio theory?

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- 2) What is an Efficient Portfolio Frontier?

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- 3) Discuss some of the limitations of Markowitz's portfolio theory.

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18.6 LET US SUM UP

This Unit explained the intricacies of portfolio management, beginning with the fundamental concept of diversification. Diversification involves spreading investments across various assets and asset classes to mitigate risk and optimize returns, offering a well-balanced approach to wealth management.

Portfolio management is the central focus of this Unit. It means overseeing and optimizing investment collections, including activities such as asset allocation, risk assessment, and performance monitoring. The primary objective is to align portfolios with investors' financial goals, risk tolerance, and timeframes, achieved through thorough analysis and proactive decision-making. The stages of portfolio management include understanding investor preferences, formulating investment policies, selecting assets, tracking performance, and making necessary adjustments. This dynamic process necessitates constant attention and adaptation to changing market conditions. The Unit also explored the relationship between risk and return, essential for investment decision-making. Investors seek to strike a balance between risk and return, with strategies ranging from active management for potentially higher returns to passive management for market-based returns and reduced risk exposure. Risks inherent in investments include diversifiable and systematic risks, each requiring distinct risk management strategies. Diversification plays a crucial role in mitigating diversifiable risks, while

systematic risks impact the overall market and require broader risk management approaches. Furthermore, the Unit explained portfolio valuation and expected returns, crucial aspects influencing investment decisions. Portfolio valuation involves assessing the value of individual assets within a portfolio, while expected returns forecast potential profits or losses, guiding portfolio selection. In this Unit, we also explored Markowitz's Portfolio Theory, a seminal contribution to portfolio management. This theory emphasizes optimizing returns while managing risk through efficient portfolio construction. Despite its merits, Markowitz's theory faces criticisms, including unrealistic assumptions and challenges in implementation.

18.7 KEY WORDS

Diversification	: The practice of spreading investments across various assets and asset classes to mitigate risk and optimize returns.
Markowitz Portfolio theory	: An investment theory based on the idea that risk-averse investors can construct portfolios to optimize or maximize expected return based on a given level of market risk, emphasizing that risk is an inherent part of higher reward
Efficient Portfolio Frontier	: A set of investment portfolios that are expected to provide the highest returns at a given level of risk.
Portfolio	: A collection of investments owned by an individual or entity, which can include various types of assets like stocks, bonds, mutual funds, and real estate.
Portfolio management	: The practice of overseeing and optimizing portfolios. It involves carefully deciding how to allocate resources, balancing risk and returns, and adapting to changes in the market.
Diversifiable risk	: It refers to firm-specific risks that affect individual stock prices rather than the entire industry or sector.
Non-Diversifiable Risk	: The risks that affect the overall market or a specific sector. It is influenced by factors that impact the entire economy.
Expected Returns	: The anticipated profit or loss an investor can reasonably foresee from an investment within a specific timeframe

18.8 SOME USEFUL BOOKS

- Strong, R. A. (2006). Portfolio construction, management and protection. Thomson/South-Western.
- Mishkin, F. S., & Eakins, S. G. (2006). Financial markets and institutions. Pearson Education India.
- Counsel, I. (2008). Portfolio Management: Theory & Practice.
- Dicksler, James L. and Samuelson, Paul A., Investment Portfolio Decision-Making, London, Lexington Books, 1974.

18.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) A portfolio is essentially a collection of investments owned by an individual or entity, which can include various types of assets like stocks, bonds, mutual funds, and real estate. Stages: Understanding the goals, Creating and executing strategies, choosing asset classes and specific securities, assessing and tracking the portfolio's performance and analyzing the portfolio's outcomes
- 2) Consistent Risk Measurement, guidance for Strategic Planning, increased Transparency in Decision Making, understanding value creation, identifying improvement areas.

Check Your Progress 2

- 1) Refer to Sub-sections 18.3.2 and 18.3.3
- 2) Diversifiable risk, also known as unsystematic risk (unique risks), refers to firm-specific risks that affect individual stock prices rather than the entire industry or sector. Systematic risks, or non-diversifiable risks, are the risks that affect the overall market or a specific sector.

Check Your Progress 3

- 1) Risk-averse investors can construct portfolios to optimize or maximize expected return based on a given level of market risk. There is a direct relationship between the variability of returns and risk: the higher the variability, the higher the risk. Return can be maximised through a combination of securities whose total variability is lower.
- 2) An efficient portfolio is one where no higher expected return can be attained without accepting a higher level of risk. Conversely, it's a

portfolio where no greater certainty of returns can be obtained without sacrificing expected returns.

- 3) Unrealistic assumptions, Complexity and Practicality, Sensitivity to input data, neglect of transaction costs and taxes, single-period analysis, ignorance of Behavioural Finance insights, inability to capture tail risks.

